SharedPtr
#nh_
#node_
SmaccStatus
#status_msg_
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
#orthogonals_
shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >
#stateMachineInfo_
recursive_mutex
-eventQueueMutex_ -m_mutex_
atomic< StateMachineInternal Action >
-stateMachineCurrentAction
map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >
-stateCallbackConnections
map< std::string, std ::pair< std::function < std::string()>, boost ::any > >
-globalData_
vector< smacc2_msgs ::msg::SmaccTransitionLogEntry >
-transitionLogHistory_
SMRunMode
-runMode_
int64_t
-stateSeqCounter_
SharedPtr
#timer_
vector< smacc2::ISmaccState * >
#currentState_
shared_ptr< smacc2 ::introspection::SmaccState Info >
#currentStateInfo_
SharedPtr
#eventsLogPub_ #stateMachinePub_ #stateMachineStatusPub_ #transitionHistoryService_ #transitionLogPub_
-statusPub_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
vector< std::shared _ptr< smacc2::introspection ::SmaccStateInfo > >
+children_
type_info *
+tid_
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccEventGenerator Info > > >
+eventGeneratorsInfo
std::enable_shared _from_this< SmaccStateInfo >
map< const std::type _info *, std::vector < smacc2::introspection ::ClientBehaviorInfoEntry > >
+staticBehaviorInfo
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccStateReactor Info > > >
+stateReactorsInfo
weak_ptr< smacc2::introspection ::SmaccStateMachineInfo >
+stateMachine_
weak_ptr< smacc2::introspection ::SmaccStateInfo >
+parentState_
vector< smacc2::introspection ::SmaccTransitionInfo >
+transitions_
int
+depth_ +stateIndex_
string
-demangledStateName +fullStateName
atomic< bool >
vector< std::shared _ptr< smacc2::ISmaccClient > >
#clients_
vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >
-clientBehaviors_
mutex
-mutex_
-stateMachine_
smacc2::introspection ::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
-orthogonal_
-stateMachine_
vector< smacc2::ISmaccUpdatable * >
-updatableClients_
vector< std::vector < smacc2::ISmaccUpdatable * > >
-updatableStateElements_
atomic< int64_t >
-lastState_
double
-loop_rate_hz
SmaccFifoScheduler *
-scheduler_
processor_handle
-processorHandle_
thread
-signalDetectorThread_
ExecutionModel
-executionModel_
-end_ -initialized_ -rosInitialized_
-signalDetector_
-smaccStateMachine_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature >::result_type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >
-stateMachine_
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
#stateMachine_
#components_
shared_ptr< rclcpp ::Logger >
vector< std::shared _ptr< smacc2::StateReactor > >
vector< std::shared _ptr< smacc2::SmaccEventGenerator > >
#stateInfo_
#logger_
#stateReactors_
#eventGenerators_
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
-isShutdownRequested_
#parentState_
-heartbeat_interval _ms_ -port_ -slave_id_
-ip_address_
-currentOrthogonal
-mutex_
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
boost::signals2::signal < void(const std::string &), optional_last_value < typename boost::function _traits< void(const std::string &)>::result_type >, int, std ::less< int >, function< void(const std::string &)>, typename extended _signature< function_traits< void(const std::string &)>::arity, void(const std ::string &)>::function_type, boost::signals2::mutex >
optional< rclcpp::Time >
optional< rclcpp::Duration >
boost::signals2::signal < void(int channel, bool state), optional_last_value < typename boost::function _traits< void(int channel, bool state)>::result_type >, int, std ::less< int >, function< void(int channel, bool state)>, typename extended _signature< function_traits< void(int channel, bool state)>::arity, void(int channel, bool state)>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal < void(int channel, const std::string &error), optional _last_value< typename boost:: function_traits< void(int channel, const std::string &error)>::result _type >, int, std::less< int >, function < void(int channel, const std::string &error) >, typename extended_signature< function_traits < void(int channel, const std::string &error)> ::arity, void(int channel, const std::string &error) >::function_type, boost::signals2::mutex >
boost::signals2::signal < void(uint8_t states) , optional_last_value< typename boost::function _traits< void(uint8_t states) >::result_type >, int, std:: less< int >, function< void(uint8 _t states)>, typename extended_signature < function_traits< void(uint8_t states) >::arity, void(uint8_t states)>::function _type, boost::signals2::mutex >
+COIL_BASE_ADDRESS +NUM_CHANNELS
-currentState
< void()>
-lastUpdate_
-periodDuration_
< void(const std::string &)>
< void(int channel, bool state)>
< void(int channel, const std::string &error)>
< void(uint8_t states)>
#owner_
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
optional< std::future < int > >
smacc2::SmaccSignal < void()>
function< void()>
smacc2::SmaccSignal < void(const std::string &)>
bool
modbus_t *
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
smacc2::SmaccSignal < void(int channel, bool state)>
smacc2::SmaccSignal < void(int channel, const std::string &error)>
smacc2::SmaccSignal < void(uint8_t states)>
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
-onEntryThread_ -onExitThread_
-onFailure_ -onFinished_ -onSuccess_
-postFailureEventFn_ -postFinishEventFn_ -postSuccessEventFn_
+onConnectionLost_ +onConnectionRestored_
-postConnectionLostEvent_ -postConnectionRestored Event_
+onConnectionError_
-connect_on_init_ -connected_
-ctx_
smacc2::SmaccAsyncClient Behavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
cl_modbus_tcp_relay ::CpModbusConnection
+ CpModbusConnection()
+ ~CpModbusConnection()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ connect()
+ disconnect()
+ reconnect()
+ isConnected()
+ getContext()
+ getMutex()
+ getIpAddress()
+ getPort()
+ getSlaveId()
+ onConnectionLost()
+ onConnectionRestored()
+ onConnectionError()
# update()
- declareAndLoadParam()
-postWriteFailureEvent_ -postWriteSuccessEvent_
+onWriteSuccess_
+onWriteFailure_
+onReadComplete_
-connectionComponent_
cl_modbus_tcp_relay ::CpModbusRelay
+ CpModbusRelay()
+ ~CpModbusRelay()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ writeCoil()
+ readCoil()
+ writeAllCoils()
+ writeAllCoils()
+ readAllCoils()
+ onWriteSuccess()
+ onWriteFailure()
+ onReadComplete()
- channelToAddress()
- isValidChannel()
-channel_
-connectionComponent_
-relayComponent_
cl_modbus_tcp_relay ::ClModbusTcpRelay
+ ClModbusTcpRelay()
+ ~ClModbusTcpRelay()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ getConnectionComponent()
+ getRelayComponent()
-relayComponent_
-client_
cl_modbus_tcp_relay ::CbRelayOff
+ CbRelayOff()
+ ~CbRelayOff()
+ onStateOrthogonalAllocation()
+ onEntry()